Optimization Model for Allocating Resources for Highway Safety Improvement at Urban Intersections

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Abstract

The authors present a procedure for allocating resources for implementing safety improvement alternatives at urban intersections over a multi-year planning horizon. The procedure, based upon optimization techniques, attempts to maximize benefits-measured in dollars saved by reducing crashes of different severity categories, subject to budgetary and other constraints. It is presented in two parts (1) a Base case including the objective function and a set of mandatory constraints, and (2) additional policy constraints / special features that can be separately incorporated to the Base case.

Demonstration of the procedure is presented on intersections in the Detroit metropolitan region, where economic losses resulting from traffic crashes at intersections are estimated to exceed $4 billion annually. The proposed model can allocate resources for safety improvement alternatives over a planning horizon, given a number of independent locations and a number of mutually exclusive alternatives at each location. The policy constraints provide the analyst the flexibility of adding equity, urgency, and other features to the Base case. Integer programming technique is applied to solve the demonstration problem.

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Introduction

In the year 2009, there were a total of 33,808 fatalities and 2.2 million injuries from highway crashes in the United States (U.S). The economic loss from these traffic crashes resulting in fatalities, injuries and property damages is estimated $230 billion per year nationally. Approximately, 40 percent of these losses are attributable to urban highways; and 75 percent of these crashes occur at intersections controlled by traffic signals or stop signs (NHTSA 2011).

Clearly, urban intersections are key locations of traffic crashes that contribute significantly to the economic losses nationally. State Departments of Transportation (DOT's), along with the U.S Department of Transportation (USDOT) spend billions of dollars annually for safety improvement programs at urban intersections. In the state of Michigan (the 'subject' state used in this paper), the economic losses resulting from highway crashes are estimated at $10 billion annually. The cost of highway crashes at urban intersections in the Detroit metropolitan region (the 'subject' region used for the paper), is estimated to exceed $4 billion annually (Khasnabis et al. 2006a).

The seven county area in Southeast Michigan with a population base of 4.5 million, has more than 25,000 intersections on its state trunkline system that constitutes the region's most traveled corridors. Of these, approximately 1,150 intersections experience more than three crashes per year and 450 intersections experience more than 10 crashes per year (based on the analysis of three years' crash data between the years 2002 and 2005) (Khasnabis et al. 2006a). Assuming 10 crashes per year as the threshold value, it would be desirable to undertake safety improvement alternatives at these intersections and to reduce economic losses. Since availability

of funds is a major factor, there is a need to reduce the number of intersections to a manageable size for any safety improvement program to be implemented over a planning horizon.

Study Objectives

The objectives of this study are three-fold:

- To present the development of an optimization model to allocate resources for safety improvement among mutually exclusive, location-specific alternatives when a number of independent locations exist.
- To improve the capability of the model by incorporating additional constraints and/or relaxation features to reflect different policy and practical considerations.
- To demonstrate the application of the model, along with the modifications on an example problem depicting reality to the extent possible.

Problem Statement

The discussion presented above encompasses three major problem areas,

1. Selections of hazardous locations, (candidate locations) where safety improvements are warranted.
2. Development of mutually exclusive alternatives (each alternative consisting of single or multiple countermeasures) at each candidate locations.
3. Allocation of resources among the independent candidate locations in conformance to budgetary and other constraints.

The combination of the above three steps in effect constitutes the safety management process undertaken by most states, and is often referred to as the hazard elimination program. The focus of this paper is in the third area, allocation of resources among mutually exclusive, location-specific alternatives and among a number of independent locations as a part of the agency's safety improvement strategy. The other two problem areas, identification of locations and development of countermeasure are mentioned earlier to provide a broader perspective. These steps require a number of comprehensive studies of crash data along with geometric and traffic conditions. As mentioned earlier, only a fraction of locations initially identified as hazardous are actually selected for implementation of safety projects because of funding limitations. These are discussed extensively in the literature (Tarko and Kanodia 2004; Hauer 1996; Deacon et al. 1975; Craig et al. 2007; Lambert et al. 2003; Cook et al. 2001; Hossain, and Muromachi 2011). The problem investigated in this paper can be articulated as:

"*Given a number of independent locations, and a number of mutually exclusive location-specific alternatives, how to allocate resources to implement alternatives at different locations over a defined planning horizon to maximize benefits within budgetary and other constraints?*"

Literature Review

There is a large body of literature on the topic of resource allocation/optimization techniques that span across such diverse areas as operations research, manufacturing, management, finance, and transportation. Optimization usually involves the maximization or minimization of an objective function comprising a set of decision variables, subject to various constraints (Bierman et al. 1997; Hillier and Liberman 2005). The constraints are designed to reflect limitations imposed by

practical and / or policy considerations, expressed in the form of inequalities or equalities. Different optimization techniques such as linear programming (LP), integer programming (IP), nonlinear programming, and dynamic programming have been used to allocate resources on various engineering and management problems (Rau 1996; Wolsey and Nemhauser 1999). The authors provide below a brief overview of recent optimization studies for allocating resources for highway safety improvement programs. The review presented is by no means a comprehensive one; rather it is designed to capture a cross-section of studies conducted on this subject during the last fifteen years.

One of the earlier studies on the use of optimization for allocating resources for safety alternatives attempted to incorporate uncertainty in estimates of accident frequency in decision making process (Persaud and Kazakov 1994). The algorithm used attempts to maximize the net benefit of a treatment at a location, where the net benefit is defined as the difference between safety benefits and costs. Further, the net benefit can be adjusted by the user through a threshold value. Researchers have also attempted to minimize total number of crashes in the study area over a multi-year analysis period, using integer programming techniques within the constraints of a specified budget (Pal and Sinha 1998). The multi-year feature provides the analyst the flexibility of utilizing the carry-over funds to subsequent years. The authors of this paper used this approach in the model presented.

The concept of Interactive Multi-objective Resource Allocation (IMRA) model has been used by researchers for selecting highway safety improvement alternatives (Chowdhury et al. 2000). The multi-objective feature of the model is designed to keep the stated (and often conflicting) objectives in their respective units and to provide a set of logical and feasible

solutions, rather than a single solution. Other researchers have used mixed integer programming techniques, based on branch and bound algorithm to allocate funds for highway safety improvements (Melachrinoudis and Kozanidis 2002). They used traditional binary variables to represent specific points along highways for intersection improvements and continuous variables for longitudinal highway segments for pavement resurfacing of alternatives.

As a part of NCHRP 486, a Resurfacing Safety Resource Allocation (RSRAP) model was developed by Midwest Research Institute in 2003 to allocate funds for pavement resurfacing (Harwood et al. 2003a). Integer programming techniques were used to maximize total benefits on a system-wide basis, as opposed to a location-specific basis, measured as the present value of safety benefits, subject to budget and other constraints. A detailed documentation of the procedure with a case study application showed that RSRAP model has the capability to use location-specific geometric data to select the optimal set of improvement alternatives for the study area as the final product (Harwood et al. 2003b). A separate study conducted an independent testing of RSRAP model on sections of Oregon DOT's highway network, and concluded that RSRAP can be effectively used by the state to select various improvements on pavement preservation alternatives (Grille et al. 2005).

The concept of optimization has been used to allocate resources for problems related to rehabilitation, renovation and upgrading of buildings (Shohet and Perelstein 2004). The model developed may be implemented either to maximize benefits within a budget constraint, or to minimize cost while maintaining the performance of the buildings. Even though the topic of this paper is renovation of buildings as opposed to highway safety, this paper is included in the

literature review, as it uses the concept of relative importance/urgency of alternatives in allocating resources. The authors of this paper adapted this concept to the current problem.

Optimization techniques have also been used to allocate resources for programs to reduce alcohol-related crashes (Kar and Datta 2004). They used linear programming techniques to maximize savings resulting from crash reduction in the state of Michigan. Similarly optimization has been used to select safety and operational improvement on highway networks (Banihashemi 2007). The model attempts to minimize crash and delay cost in conjunction with crash prediction models adapted from the Interactive Highway Safety Design Model (IHSDM) software within the constraint of a given budget (IHSDM 2010).

The literature review presented above shows that the topic of resource allocation for safety improvement alternatives has received significant research attention during last fifteen years. Within the general framework of optimization approach, researchers have used different model formulations and different solution techniques to address their respective issue. Models formulated include minimization of crashes, maximization of savings in crashes, maximization of benefits measured in dollars; and the techniques used include Linear Programming, Integer Programming, Dynamic Programming, etc. Most of the papers reviewed allocated resources for one year; only a limited few attempted multi-year allocation with a planning horizon in mind. Different researchers have treated constraints differently to reflect various policy and practical considerations.

In this paper, the authors present an approach to optimize the safety benefits in a given area by maximizing the dollar value of the crashes saved at intersections each year over a multi-year planning horizon. There is a lack of consensus among researchers regarding the use of a

dollar value of the crashes saved as a measure of benefits for resource allocation purposes. The dollar value, being the most common denominator, is the preferred measure of many researchers. Benefits measured in dollars, are however likely to be highly influenced by fatal crashes, as its unit value is significantly higher than those of all other crashes. Since fatal crash is a rare event at the individual intersection level, other researchers argue that the use of dollar value would, in effect, result in a rare event significantly affecting funding decisions.

The authors' primary reasons for using the dollar value are three fold: (1) The dollar value is the most common/measurable denominator used in the investment decisions, (2) Very few of the papers reviewed used the dollar value as a measure of benefit, and (3) Funding priorities can also be influenced by proper treatment of constraints. The example illustrated in this paper clearly demonstrates that key constraints (e.g. "mutually exclusive", "carry-over" effect, county constraint, and urgency constraint) can act as "buffers" against providing undue priorities to the locations with the rare event, i.e. fatal crash, thereby bringing about a more equitable allocation of resources.

Model Formulation

The model is presented in three steps. First, key definitions and notations are introduced; second, a Base case is presented which consists of an objective function and a set of mandatory constraints; third additional constraints reflecting a number of policy options are introduced. Notations used in the formulation are explained below.

Variables	Explanation
b_n	Allocated budget ($) in the analysis year *n*
$c^{n,f}$	Cost of fatal crash (*f*) in year *n*

$c^{n,m}$	Cost of injury crash (m) in year n
$c^{n,p}$	Cost of property damage (p) crash in year n
f_i^n	Expected number of fatal crashes for location i in analysis period n
I_ε	Total number of locations for the county ψ_ε, where $\psi_\varepsilon \in \{\psi - \psi_\delta\}$
I_δ	Total number of locations belong to the county ψ_δ
k_j	The number of years an alternative j is in operation and effective after installation and before the end of service life
l_j	Service life of the alternative j
m_i^n	Expected number of injury crashes, m, for location i in analysis period n
$o_{i,j}^n$	Operation and maintenance cost for alternative j implemented in location i in the analysis year n
p_i^n	Expected number of property damage only crashes, for location i in analysis period n
$r_{i,j}^p$	Crash reduction factor for property damage, p, alternative j chosen for location i
$r_{i,j}^f$	Crash reduction factor for fatality f, alternative j chosen for location i
$r_{i,j}^m$	Crash reduction factor for injury m, alternative j chosen for location i
s_i^n	Relative urgency score for location i in the analysis year n
$x_{i,j}^{k_j,n}$	An auxiliary binary decision variable exists only for an alternative after first year of implementation but before the service life = 1 when a new alternative j is implemented at location i for the analysis year n, and is effective for k_j years after installation, where $1 < k_j \leq l_j$ = 0 Otherwise
$y_{i,j}^{k_j,n}$	An auxiliary binary decision variable for a new alternative implementation = 1 when a new alternative j is implemented at location i for the analysis year n, the alternative is effective for the first year of installation, where k_j=1 = 0 Otherwise
$z_{i,j}^{k_j,n}$	Binary decision variable. =1 when an alternative j is chosen for location i for the analysis year n, the alternative is effective for k_j years after installation = 0 Otherwise
$\hat{z}_{i,j}^{k_j,n}$	Binary decision variable. =1 when an alternative j is suggested for location i for the analysis year n, the alternative is effective for k_j years after installation (before allocation) =0, when an alternative j is not suitable for location i

$\gamma_{i,j}^{k_j,n}$	Number of alternatives allocated to location *i* in the year *n* (where type of alternative *j* may vary), and the alternatives remain effective for $k_j$years after installation till the end of service life
$\gamma_{i,j}^{\psi_\delta,n}$	Number of alternatives (multiples of *j*) allocated to county ψ_δ (consisting of multiple locations, *i*) in the analysis year *n*
$\gamma_{i,j}^{\psi,n}$	Number of alternatives (multiples of *j*) allocated to the subject county ψ (consisting of multiple locations, *i*) in the analysis year *n*
δ	Subscript used for a county
θ	A threshold value set by the decision maker for equitable allocation of resources to counties
ϑ^n	A threshold value representing a measure of relative score for year *n*
$\pi_{i,j}^n$	Capital cost for alternative *j* implemented in location *i* in the analysis year *n*
ψ_δ	Representing a specific county δ from a set
ω^f	Weighting factor for fatal crash
ω^m	Weighting factor for injury crash
i	Location in the study area
I	Total number of locations
I'	A subset of *I*
j	Alternative proposed to be have potential for crash reduction
J	Total number of alternatives
j'	Alternative selected for installation in addition to an existing alternative *j* already in place for location *i* in the year n. *j'* is a subset of *J* alternatives
n	Planning period under consideration
N	Total planning period
Z	Objective function, dollar benefit of crashes saved for the analysis period *n*

Base case

In the proposed model, the objective is to maximize the benefits Z derived from crashes saved for a set of locations upon implementation of alternatives for the proposed planning period of N years. The model is formulated as follows:

Maximize

$$Z = \sum_{n=1}^{N}\sum_{j=1}^{J}\sum_{i=1}^{I}\sum_{k_j=1}^{l_j}\left[f_i^n r_{i,j}^f c^f + m_i^n r_{i,j}^m c^m + p_i^n r_{i,j}^p c^p\right] z_{i,j}^{k_j,n} \tag{1}$$

subject to

$$\sum_{n=1}^{N}\sum_{j=1}^{J}\sum_{i=1}^{I}\sum_{k_j=1}^{l_j}\left[\pi_{i,j}^n y_{i,j}^{k_j,n} + o_{i,j}^n x_{i,j}^{k_j,n}\right] \leq \sum_{n=1}^{N} b_n \tag{2}$$

$$\sum_{j=1}^{J}\sum_{k_j=1}^{l_j} z_{i,j}^{k_j,n} \leq \sum_{j=1}^{J}\sum_{k_j=1}^{l_j} \hat{z}_{i,j}^{k_j,n} \tag{3}$$

$$\sum_{j=1}^{J}\sum_{k_j=1}^{l_j} z_{i,j}^{k_j,n} \leq \sum_{j=1}^{J}\sum_{k_j=1}^{l_j} \gamma_{i,j}^{k_j,n} \tag{4}$$

Where,

$$\sum_{j}^{J} \gamma_{i,j}^{k_j,n} \leq \begin{cases} \sum_{j}^{J} y_{i,j}^{k_j,n} + \sum_{j}^{J} x_{i,j}^{k_j,n} & ,\forall\, k_j > 0 \\ 0, & Otherwise \end{cases} \tag{5}$$

$$\sum_{j}^{J} y_{i,j}^{k_j,n} \leq \begin{cases} 1, & k_j = 1 \\ 0, Otherwise \end{cases} \tag{6}$$

$$\sum_{j}^{J} x_{i,j}^{k_j,n} \leq \begin{cases} 1, & 1 < k_j \leq l_j \\ 0, Otherwise \end{cases} \tag{7}$$

$$z_{i,j}^{k_j,n}, \gamma_{i,j}^{n}, x_{i,j}^{k_j,n}, y_{i,j}^{k_j,n} \geq 0, \forall\ i,j,\ k_j \tag{8}$$

In expression (1) the total benefits is measured in terms of dollars from savings in fatal, injury and property damage only (PDO) crashes. For example the first term ($f_i^n r_{i,j}^f c^f$) in expression (1) estimates the benefits received from savings in fatal crash; where f_i^n is the expected number of fatal crashes for location *i* in analysis period *n*; $r_{i,j}^p$ is the crash reduction factor for property damage, *p*, for alternative *j* chosen for location *I*; and $c^{n,f}$ is the cost of fatal crash (*f*) in year *n*. Similarly $m_i^n r_{i,j}^m c^m$ accounts for savings from injury crash and $p_i^n r_{i,j}^p c^p$ for savings from PDO. In expression (1) $z_{i,j}^{k_j,n}$ is a decision variable which takes the value 1 when an alternative *j* is chosen for location *i* for the analysis year *n*, and the alternative is effective for k_j years after installation; and 0 otherwise. Expression (2) is a budget constraint, and it ensures that the sum total of capital investment and operation and maintenance (O&M) cost should not exceed the total budget in the planning period, though there is a flexibility of expenditure between the years in the planning period. Such flexibility in expenditure between years within a planning period can be incorporated into the procedure through a Planning Budget Model (PBM) as applied in transit resource allocation (Mathew et al. 2010). PBM can be defined as a single budget considered for the entire planning period and is based on the assumption that the agency has the flexibility of borrowing monies from subsequent years' allocation. In expression (2),

$\pi_{i,j}^{n} y_{i,j}^{k_j,n}$ represents the capital cost of implementation of an alternative *j* for location *i* in year *n*; where $\pi_{i,j}^{n}$ is the capital cost for alternative *j* implemented in location *i* in the analysis year *n*, and $y_{i,j}^{k_j,n}$ is an auxiliary binary decision variable for a new alternative implementation and takes the value 1 when a new alternative *j* is implemented at location *i* for the analysis year *n*, and the alternative is effective for the first year of installation (where k_j=1); and 0 otherwise. Similarly, the term $o_{i,j}^{n} x_{i,j}^{k_j,n}$ in expression (2) accounts for O&M cost; where $o_{i,j}^{n}$ is the O&M is cost for alternative *j* implemented in location *i* in the analysis year *n*, and $x_{i,j}^{k_j,n}$ is an auxiliary binary decision variable that exists only for an alternative after first year of implementation but before the service life ($1 < k_j \leq l_j$). The total budget in expression (2) is represented as $\sum_{n=1}^{N} b_n$, where b_n is the budget available for year *n*. It should be noted that there is a flexibility of spending in individual years within the planning period.

Expression (3) ensures that $z_{i,j}^{k_j,n}$, an alternative *j*, for location *i*, for year *n* (and effective for k_j years after installation) is chosen from a set of pre-suggested alternatives ($\hat{z}_{i,j}^{k_j,n}$) for the analysis year *n*. Based on engineering design, the suggested alternatives tend to be location-specific. Expression (4) denotes that each location can receive only number of alternatives ($\gamma_{i,j}^{k_j,n}$) for the analysis year *n*, pre-specified by the planning agency. When the alternatives are mutually exclusive, the maximum value of $\gamma_{i,j}^{k_j,n}$ is equal to one for each location (for the Base case), where $\gamma_{i,j}^{k_j,n}$ number of alternatives are allocated to location *i* for year *n* (where type of alternative *j* may vary), the alternatives remain effective after installation for the remainder of service life. The benefits of alternatives for *k* years after installation (where $k_j \leq l_j$) is included in the benefits expressed in the objective function. Expression (5) is a definitional constraint which

takes binary values (one or zero) based upon allocation of an alternative *j* at location *i*. In expression (5), when a new alternative is implemented, $y_{i,j}^{k_j,n}$ is equal to one for the first year of implementation (k_j=1) and $x_{i,j}^{k_j,n}$ is equal to one for the remainder of service life ($1 < k_j \leq l_j$). It should be noted that the values of $y_{i,j}^{k_j,n}$ and $x_{i,j}^{k_j,n}$ cannot be 1 simultaneously. Expression (5) ensures that when an alternative is implemented, no other alternative is funded at that location during its service life. Thus, the "mutually exclusive" characteristic of the model ensures two features;

- *Feature 1*: A location can receive only one alternative in a given year.

- *Feature 2:* A location, that has the carry-over effect from an alternative implemented in previous years, may not receive any funds during the service life of the alternative. (Note: Depending on the availability of funds, and other factors, it may be necessary to relax the second feature. It is discussed under relaxation of carry-over feature later.

Expression (6) is a definitional constraint which denotes that, $y_{i,j}^{k_j,n}$, a binary variable indicator to be multiplied with the capital cost of an alternative *j* selected for location *i* in the year *n*. $y_{i,j}^{k_j,n}$ is equal to one for the first year (k_j=1), and zero for the remainder of the service life of the alternative. Expression (7) is a definitional constraint (similar to expression (6)) which denotes that a binary variable ($x_{i,j}^{k_j,n}$) indicator to be multiplied with operation and maintenance cost of an alternative *j* selected for location *i* in the year *n,* $x_{i,j}^{k_j,n}$ is equal to zero for the first year and one after the first year till the end of the service life ($1 < k_j \leq l_j$). Expression (8) is a non negativity constraint of the decision variable $z_{i,j}^{k_j,n}$, and the auxiliary decision variables.

Policy Constraints and Other Factors

A set of additional policy constraints are presented that can be added either singly or in combination to the Base case at the discretion of the user.

<u>*Relative urgency constraint case*</u>

This case expresses the relative urgency constraint with which a safety improvement is to be applied to a location. Higher priorities are assigned to locations with crashes of higher severity. Relative urgency of a location can be expressed by means of a relative score (expression 9), which consists of weighting factors for fatal and injury crashes (expression 10). The relative score (s_i^n) of a location can be determined as;

$$s_i^n = \left[\omega^f f_i^n + \omega^m m_i^n + p_i^n\right] \tag{9}$$

Where,

$$\omega^f = \frac{c^f}{c^p}, and\ \omega^m = \frac{c^m}{c^p} \tag{10}$$

A threshold value of the relative urgency ϑ^n is estimated as the mean of relative scores of all the locations in year *n*, i.e. $\frac{\sum_{i=1}^{I} s_i^n}{\sum_{i=1}^{I} i}$. A binary variable is defined based on the threshold value ϑ^n for each location to incorporate its relative urgency, which is defined as follows (expression 11):

$$\gamma_{i,j}^{k_j,n} = \begin{cases} 1, & \vartheta^n \geq s_i^n \\ 0, & Otherwise \end{cases} \tag{11}$$

Expression (11) suggests that only locations regarded as "relatively urgent" are eligible for resource allocation. The threshold value can be any desired value chosen by the planner. In this paper, the mean value ($\vartheta^n = \frac{\sum_{i=1}^{I} s_i^n}{\sum_{i=1}^{I} i}$) is considered as the threshold value. The relative urgency constraint is designed to ensure that locations affected by higher fatal and injury (more severe) crashes receive priority over other locations.

<u>*County constraint case*</u>

Each location in the study area is associated with a specific county. A policy constraint may be designed to ensure that each county receives an equitable distribution of funds. This condition can be achieved by the following constraint:

$$\sum_{n=1}^{N}\sum_{j=1}^{J}\sum_{i=1}^{I_\delta}\gamma_{i,j}^{\psi_\delta,n} \leq \theta \sum_{n=1}^{N}\sum_{j=1}^{J}\sum_{i=1}^{I_\varepsilon}\gamma_{i,j}^{\psi,n}, \forall\, \psi_\varepsilon \in \{\psi - \psi_\delta\} \tag{12}$$

where,

$$\theta > 1 \tag{13}$$

$$\psi_\delta = \sum_{i=1}^{I_\delta} i \tag{14}$$

Expression (12) ensures that a county cannot receive *j* number of alternatives unless all other counties have received a minimum number of alternatives (defined as a threshold value θ) set by the decision maker. Expression (13) is a definitional constraint for the threshold value that can be greater than one and Expression (14) ensures that each county includes a number of locations (*i*) which is a subset of the total sample of intersections (*I*). The county constraint may

be necessary if equitable distribution of alternatives is not achieved in the Base case. The threshold value (θ) as a constraint ensures equitable distribution of alternatives among counties.

Relaxation of carry-over feature case

The Base case, Relative urgency case, and County constraint case are based upon allocation of alternatives which are mutually exclusive in nature. High frequency and high severity locations require installation of multiple alternatives in a given year to maximize the benefits. If there are a number of high frequency and high severity locations, it may be necessary to relax the mutually exclusive constraint, thereby allowing the installation of different alternatives during the service life of an existing alternative at a specific location. This feature is termed as "carry-over" in the remainder of the paper. Thus, if alternative j is installed in the year n for location i, it has a carry-over effect in the year $n+1$ for location i, and for subsequent years during the service life of the alternative. Relaxation of the carry-over feature allows installation of another alternative (j') in year $n+1$, while the effect of alternative j is carried over from year n for location i. The relaxation may take on different forms, and will depend upon the specific case. In this paper, the carry-over feature is described as follows;

- A location that has a carry-over effect from an alternative implemented from previous year(s), may receive another alternative.
- However, locations with carry-over effects from two alternatives may not receive another alternative on the year in question.

The objective of this option is to maximize total benefits by allocating a new alternative when a different alternative is already in place. Relaxation of the "carry-over feature" constraint can be implemented by expression 15, and 16;

$$\gamma_{i,j}^{k_j,n} = \begin{cases} y_{i,j}^{k_j,n} + y_{i,j'}^{k_j,n}, & k_j = 1 \\ x_{i,j}^{k_j+1,n} + x_{i,j'}^{k_j+1,n}, & 1 < k_j \leq l_j \end{cases} \tag{15}$$

$$y_{i,j}^{k_j,n} = \begin{cases} 0, \forall\, x_{i,j}^{k_j+1,n} = 1 \\ 1, Otherwise \end{cases} \tag{16}$$

Expression (15) suggests that the carry-over effect is relaxed, thereby installing multiple alternatives in one year, or installing a new alternative while the effect of another alternative installed earlier is still in place. Expression (16) suggests that a new alternative $y_{i,j}^{k_j,n}$ (of similar nature) cannot be installed if another alternative is already in place for location *i* in the year *n*, within k_j number of years after installation till the end of its service life. The combined CRF for locations with multiple allocations of projects will follow the regular practice of total $CRF_t = 1 - \{(1 - CRF_1)(1 - CRF_2) \ldots . (1 - CRF_n)\}$.

Model Structure

A model structure for the proposed methodology is shown in Figure 1 as a step by step procedure. The first step of the design is to collect intersection specific crash data. A detailed information on crash severity (such as number of fatality, injury and property damage only), and types (such as rear-end, angle, side-swipe, head-on, and other) is essential in the design of location-specific countermeasures (or their combination) along with annual budgets. Expressions 1 through 8 can be applied to derive desired numbers for the Base case. Expressions 9 through

16 can be applied to incorporate policy constraints. The suitable model results then can be considered for application.

Model Application Results

The crash resource allocation model is solved by Integer Programming with branch and bound algorithm using Premium Solver Platform. (PSP 2010a, and PSP2010b). In the case study presented, an initial annual budget of $645,000. The future year budgets are assumed to increase by six percent every alternate year over a five year planning horizon. The rationale behind selecting the above initial budget is discussed in the next section. Information on factors that need to be considered from year to year for all the proposed models: mutually exclusive feature, carry-over factor[1], and year end surplus are tracked internally within the model. The model is applied to an example problem depicting reality to the extent possible to ensure a connection between the proposed process and its application / practice. An analysis period of five years is assumed in the example demonstration.

The annual savings measured in monetary terms from the reduction in number of crashes is termed as "benefit", and the savings over the five year planning period is termed as "total benefit". These two terms are used in the following sections as a measure of the monetary savings from reduction in crashes. Surplus is defined as difference between available budget and the amount committed for implementation of alternatives. The terms annual surplus and total surplus are used in the remainder of the paper for unused budget for annual and planning period respectively.

[1] An alternative installed for the first year remains effective for the remainder of its service life.

Problem Description

The resource allocation model for highway safety improvements is applied to a set of 30 intersections in the Southeast Michigan region comprising Wayne, Oakland, and Macomb counties. These three counties represent what is commonly referred to as the Detroit metropolitan area, with the city of Detroit being located within Wayne county. The Detroit metropolitan area is a part of the Southeast Michigan region. These 30 intersections selected for the example demonstration consist of 10 highest crash frequency locations from each of the three counties and are a sub-set of 25,000 intersections (discussed in the Introduction section) in the region. The authors realize the need for a model to analyze a larger number of intersections recognizing that there are far more than 30 hazardous locations in the region. But to avoid the complexity of analyzing probable causes and designing location-specific appropriate countermeasures, a subset of 30 locations is selected for demonstration purposes. An implied assumption in limiting the study to intersections is that there is a targeted budget for the treatment of these type of locations.

Annualized crash data (over a 10-year period) for 10 intersections from each county, for a total of 30, compiled from the website of the Southeast Michigan Council of Governments (SEMCOG) is presented in Table 1 (SEMCOG 2008). These intersections are listed in decreasing order of total crashes, and represent the demonstration platform for the proposed resource allocation model. Costs for fatal, injury, and property damage crashes were assumed as $1,200,000, $55,000 and $8,200 respectively per National Safety Council (NSC 2010).

Information on crash types, available from state police (not shown in Table 1) are typically used to design appropriate countermeasures for each location, as they provide significant insights

to the probable causes of crashes and hence augment the design of countermeasures. The term relative score is a composite measure of frequency and severity of crashes at a location reflecting the need for attention for alternative implementation, as described in equation (9). Weighting factors for fatal and injury crashes are calculated as 146.3 (i.e.$1,200,000/$8,200) and 6.7 (i.e.$55,000/$8,200) respectively. Relative scores are used to determine the rank of each location. Thus, a location with highest total number of crashes may not get highest relative score, because of the composite measure of frequency and severity embedded in relative score estimation. The details of the relative scores and rank are discussed later in the paper under Relative urgency case. Five hypothetical safety alternatives are proposed as the countermeasures for potential reduction in crashes. Also, for demonstration purposes, it has been assumed that between three to four of the five alternatives are applicable to each of the 30 locations, as shown in Table 1. This assumption was necessary to demonstrate the "mutually exclusive" feature of the proposed resource allocation model. In reality, mutually exclusive alternatives are developed based upon engineering judgments, an analysis of the probable causes of the crashes, such that the likelihood of future crashes, (or severe injuries resulting from future crashes) is minimized.. The first row of Table 1 indicates that for location 1, alternatives I, II, III, and V are appropriate over the planning horizon. However, for a given year, these alternatives are mutually exclusive.

The capital cost of the proposed alternatives in increasing order is presented in Table 2. For the sake of simplicity, O&M costs have been assumed as 10 percent of the capital costs, and service lives for the alternatives are assumed to be generally proportional to the capital costs. Each alternative has been assumed to consist of a set of countermeasures and with hypothetical crash reduction factors (CRF) for each alternative. In reality, crash reduction factors for each countermeasure, along with its expected service life can be derived from the literature

(Khasnabis et al. 2006b, and FHWA 2007). An alternative may consist of a single or multiple countermeasures. In the latter case, CRF's associated with each countermeasure are combined following a linear function to derive a combined CRF for the said alternative (FHWA 1981). The CRF values listed in Table 2 can be assumed to be associated with each alternative (that may be a combination of countermeasures).

If minimum cost alternatives are assigned manually (without optimization) to each location, assuming that all the locations will receive at least one (the lowest cost) alternative, it is found that a total expenditure of $645,000 in the first year will generate a benefit of $1.96 million at the end of the year (measured as the worth of the crashes prevented) . For brevity, the detailed calculation for this step is not presented here. Hence, the proposed budget for the resource allocation model was assumed to be $645,000 for the first and the second year. Further, the budget was assumed to increase by six percent every alternate year over the five-year planning cycle.

Results of optimization were used to allocate $645,000 during the first year only shows that the choice of alternatives for different location is different from manual allocation (Table 3). Table 3 shows that for the one-year optimization case, alternatives II and V are selected for locations 2 and 4 respectively, while none is allocated to locations, 1, 3, and 5. The optimization procedure produces a benefit of $2.97 million (versus 1.96 million with low cost manual allocation) with the same cost of $645,000. The optimization procedure produced a different allocation of alternatives among locations to maximize the total expected benefits. However, it should be noted that not all the intersections were funded under the optimization scheme, as shown in Table 3.

Model Application

Results of optimization for the Base case (a) and with the policy constraint added, (Relative urgency case (b), County constraint case (c) and Relaxation of carry-over feature case (d)) for the five-year planning horizon are presented below. First, an intersection-level assignment of alternatives along with relevant data on costs and benefits is presented only for case (a) in Table 4. Next, results of the four cases analyzed (case (a), (b), (c), and (d)) are summarized in two Tables. Table 5 shows the financial data by year along with the assignment of alternatives for the four cases. Table 6 shows the distribution of the alternatives among the three counties in summarized form. A detailed intersection-level discussion of cases (b), (c), and (d) is not presented for brevity.

Base case (a)

The base case intersection-level assignment over the five-year planning horizon using the optimization function and the mandatory constraints are presented in Table 4. A key assumption in the optimization model is that the same alternative may be allocated to a given location more than once, only after the carry-over effect of the said alternative has expired. Table 4 only shows parts of allocation for each year for brevity. The mutually exclusive feature of the allocation process for the base year can be observed in Table 4. For example, alternative II (with a service life of two years, and a cost of $35,000) is installed in location 1 in the first year resulting in a benefit of $153,788. Benefit derived at the end of the first year amounts to $2,327,141 against a cost of $645,000 as budgeted. Hence, alternative II (or any other alternative) cannot be installed at the same location in the second year. No new alternative is assigned to location 1 in the second year. However, the carry-over effect of alternative II for location 1 requires an O&M cost of

$3,500 O&M in the second year, resulting in benefit of $133,795. Benefits received in the second year are less than those in the first year because number of crashes was reduced in the second year after installation of alternative II in the first year for location 1.. Similar allocations for different locations in the planning period are shown in Table 4.

Table 5 shows that optimization resulted in 13 new alternatives in the first year. The capital cost for implementing these alternatives is $645,000 leaving no surplus. The O&M cost is zero, as these costs are incurred one year after the alternative is implemented. The optimum benefit for the first year is computed as $2.32 million. In the second year, optimization resulted in the selection of 7 new alternatives with a capital cost of $445,000 and benefit of $3.72 million. The effect of carry-over alternatives from the previous year is also included in the estimation of the benefits derived. Similar allocations are made for five years. The benefit for the first year resulting from a single year analysis is $2.97 million (Table 3), while the first year benefit from a multi-year analysis is $2.32 million (Table 5). The difference in the benefit is simply a reflection of the fact that the model allocates resources over the five-year period optimally resulting in a greater a flexibility of investment from year to year. An analysis of one year at a time, on the other hand, is blind to availability of future funds, and may not necessarily result in maximization of total benefit over the five-year period.

Table 5 also shows that a total of 43 new alternatives are selected in the five year planning period for case (a). The total benefit achieved is worth $18.29 million at an expense of $2.95 million of capital cost and $397,500 of O&M cost, leaving a surplus of $8, 113. While the model maximizes total benefit over the five-year period, it does not guarantee that all the

locations will receive at least one alternative during the planning cycle, as this condition was not explicitly incorporated in the model formulation.

Table 6 shows that for case (a), six alternatives were allocated to Wayne county, three to Oakland county, and four to Macomb county in the first year. In the second year, one alternative was allocated to Wayne county, three to Oakland county, and three to Macomb county. The first year alternatives are carried over to the second year because of multiple year service life of alternatives. At the end of the second year, seven alternatives are allocated to Wayne county, six alternatives for Oakland county, and seven alternatives for Macomb county. Similar distribution of alternatives over counties for the five years is presented in Table 6.

Relative urgency case (b)

In the allocation for relative urgency case, locations are prioritized based on their relative scores. The threshold value for relative urgency is determined as the mean relative score of all locations. For the first year, the mean relative score (or the threshold value, ϑ^n in equation 11) is 150.22. Locations with higher relative scores (than the threshold value) receive priority in funding allocation in such a way that the total benefit is maximized, subject to budget and other constraints.

Table 5 shows that the total benefit achieved for case (b) is worth $17.91 million at an expense of $2.96 million of capital cost and $381,000 of O&M cost, leaving a surplus of $9,613. The total benefit received is higher for the case (a) ($18.29 million), when compared to case (b) ($17.91), as the addition of the relative urgency clause has the effect of further constraining the solution space. Table 6 shows that for case (b) three alternatives were selected for each county. In the second year, none was funded for Wayne county, five to Oakland county, and one to

Macomb county. At the end of the second year, three alternatives are selected for Wayne county, eight for Oakland county and four for Macomb county. Similar distribution of alternatives over the three counties for the five years is presented in Table 6 for case (b).

Tables 6 also show that in none of the two cases (a) and (b) an equitable distribution of alternatives among counties is achieved (because equity in distribution is not factor in the Base case). Such equity can however, be accomplished by stipulating that no county will receive more than two alternatives unless all other counties have received at least one (i.e. not more than twice). This is discussed in case (c).

County constraint case (c)

For incorporating the county constraint, a threshold value θ (equation 13) of two was used, implying that no county may receive more than two alternatives unless all other counties have received at least one during the planning period.. Table 5 shows that for case (c) 14 alternatives are selected at a cost of $475,000, resulting in a benefit of $1.66 million and a surplus of $170,000 in the first year. A total of 50 new alternatives are selected resulting in a total benefit of $17.52 million. The Capital and O&M costs of $3.11 million and $229,500 respectively are incurred, resulting in a total surplus of $16,113. Table 6 shows that for case (c), no county received more than two alternatives, (or multiples of thereof) unless all other counties received at least one alternative.

Relaxation of carry-over feature case (d)

Cases (a), (b), and (c) are all based upon selection of alternatives which are mutually exclusive in nature. The carry-over feature built into the mutually exclusive premise ensures that the same alternative cannot be installed at a given location during the service life of the said alternative. A

relaxation of the carry-over feature permits the installation of multiple alternatives during the service life of an existing alternative.

Table 5 shows that for case (d) a total of 10 alternatives are s for the first year, with an allocated cost and resulting benefit of $645,000 and $2.43 million respectively. The allocated cost for the first year is $645,000. In the second year, 12 alternatives are selected, and the resulting benefit is $4.72 million (including the carry-over from the first year). The capital and O&M costs are $575,000 and $64,500 respectively. The total benefit received is $19.93 million, at a capital cost of $3.13 million, and an O&M cost of $217,000. Case (d) with a surplus of $8,613results in the highest total benefit among all options considered. County-wise distribution of alternatives is not presented for case (d) for brevity, as the objective is to maximize benefit, and not to incorporate the county constraint.

Synthesis of Results

Selected output from the resource allocation model over the five year planning horizon is presented in Figure 2 with the object of assessing the "reasonableness" of the performance of the model. For the four cases analyzed, the total budget provided is the same, i.e. $3,355,613. There are however, some differences in the model output, as expected. These are summarized below:

- In all the five cases, the amount committed (inclusive of the O&M cost) is within the allocated budget (Figure 2a).
- The smallest amount committed is for the County constraint case, (resulting in the largest surplus), and the largest amount committed is for the Base case (resulting the smallest surplus) (Figure 2d).

- Total benefit (measured in terms of savings in crash cost) is maximized when the "carry-over" feature is relaxed. This scenario also results in the largest number of alternatives funded (Figure 2c).

- The overall Benefit-Cost (B/C) ratio, not discounted for any interest factor, varies from a high of 5.96 to a low of 5.25 (Figure 2f). The highest ratio is attained when the carry-over feature is relaxed, while the lowest ratio occurs with the addition of the county constraint. (Note: The benefit is presented in Figure 2 (c) and costs are presented in Figure 2(a), and 2(b). The B/C represents the ratio of total benefits to the total costs.

- The results show that if equity in funding distribution is one of the missions, the county constraint feature is a means to achieve this..

- The addition of "Relative urgency case" to the Base case results in the least desirable output, both in terms of total benefits and number of alternatives funded when compared to the Base case; even though it results in little higher total surplus (Figure 2c, and 2e). However, surplus, is not a direct, or a desired output of the model, rather it is a by-product. The addition of the relative urgency constraint has the effect increasing the alternative cost and reducing benefit as expected. The combined effect of these two phenomena is reflected in the output.

- In terms of number of alternatives funded, the Base case, and Relative urgency case (added to the Base case) produce similar results (Figure 2e). The county constraint case (added to the Base case) results in the least number, while the relaxation of carry-over constraint case results in the largest number.

- The sensitivity of the model to budgetary constraint is examined. As expected, the model is found sensitive to the small relaxations. It was found that benefits vary with changes in budget. Higher budgets result in higher benefits, and vice versa. Similarly, the number of alternatives funded increases with an increase in the amount budgeted. The amount allocated to alternatives is generally in proportion to the amount budgeted, with the provision that the sum total of the amount allocated, the O & M cost, and the surplus equals the amount budgeted. Details of the sensitivity analysis are not presented in the paper for brevity.

Overall, the model output is considered reasonable, and the trends observed followed are logical. These are reflected in various performance factors discussed above, such as: amount committed, total surplus, and number of alternatives funded, and the overall benefit-cost ratio. The whole methodology is implemented in a VBA based solver platform (PSP 2010a), on an Intel (R) Xeon (R) Core 2 Quad, 4GB memory, 2.0 GHz under Windows XP operating system. A precision value of 1.0E-6 is used to determine how closely the estimated constraints match with the given values. Each optimization run requires 30,000 iterations to find the optimal value. Each iteration requires 0.09 seconds, and one complete optimization run requires approximately 46 minutes. However, depending upon the type of problem, nature of the objective function and the constraints the computational time may differ. In addition, as the size of the problem increases, the computational time may significantly increase (Martin 2001; Kerp 2010; Zhu and Lin 2011).

Conclusions and Recommendations

In this paper, the authors present a resource allocation procedure in the form of an optimization model to implement safety improvement alternatives at urban intersections. Given a large number of independent locations (intersections) and a set of mutually exclusive alternatives at each location, the model allocates resources for implementing safety improvement alternatives at different locations in a manner that maximizes benefit by way of crashes saved every year in a multi-year planning period. The model is presented in two parts: (1) a Base case that includes the optimization function and a set of mandatory constraints (budgetary and other), and (2) a set of policy constraints / special features that can be incorporated into the Base case.

The authors demonstrate the application of the model using Integer Programming techniques on a number of intersections in the Detroit metropolitan area over a multi-year planning horizon. The demonstration is carried out initially with the Base case and then with policy constraints/special features "added on" to the base model. The demonstration platform is based upon the use of a combination of real-life and hypothetical data. The crash locations selected, along with the relevant crash data used to justify the selection of the locations are real. The data on costs, service lives, crash reduction factors, and the applicability of the alternatives to the intersections (to incorporate the "mutually exclusive" feature of the alternatives) are assumed, based upon the authors' judgment. Under ideal conditions, these information should be derived from actual data to lend more validity to the model. However, the primary purpose of this paper is to demonstrate the application of the proposed resource allocation model and the development of actual alternative data was beyond the scope of this paper.

The resource allocation model is considered robust in its formulation; and it attempts to maximize total benefit resulting from safety improvement alternatives, within a set of mandatory constraints satisfying budgetary and mathematical requirements.

- The multi-year feature allows the user to effectively utilize the year-end savings in subsequent years, thereby deriving the most benefit from the available resources.

- The incorporation of policy constraints/special features provides the analyst the flexibility of adding selectively, equity and urgency to the resource allocation problem. The case study shows that these features, when added to the Base case, result in significant changes in the allocation of resources. Further, in all the cases analyzed, the trends in the changes are logical.

- Similarly, the "Relaxation of carry-over" clause provides a means to support a larger number of alternatives over the same geographic area, without any adverse effect on the total benefit derived.

- Considering current economic situation, it may be difficult to borrow funds between years in a multi-year planning period, and hence, a planning budget model (PBM) may not be suitable for resource allocation. In such cases, an annual budget model (ABM) can be adopted. Additional constraints can be added to allow or to restrict carryover of surplus amount. An ABM simply restricts yearly spending to the respective annual budget specified. The authors have presented a comparative analysis of PBM versus ABM models on transit fleet resource allocation in a recent paper (Mathew et al. 2010). It should be noted that the use of ABM or PBM depends on discretion of the user and the

user may consider imposing a "cost of borrowing" on the amount borrowed by a discount factor, if a PBM is the desired approach.

The authors recommend additional research to explore the application of this framework to a larger number of locations. Other policy options to ensure that each location receives at least one alternative during the planning cycle can be analyzed. Budget constraint such as, "no deficit in any year during the planning cycle" can be incorporated into the analysis framework. Subject to the availability of appropriate data, and computing power, the model may be used to allocate resources among a large number of intersections in the SEMCOG area to maximize total benefit expressed in dollars saved by way of preventing crashes over a pre-specified planning horizon.

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Table 1. Summary of Study Locations: Crash Frequency, Severity, Score, Rank, and Suggested Improvements

Intersection	Major Name	Minor Name	County	Crash Frequency				Relative Score	Rank	Suggested Project Improvements				
				Total	Fatal	Injury	PDO			I	II	III	IV	V
1	Telegraph Rd	12 Mile Rd W	Oakland	92.5	0	16.1	76.4	184.27	6	1	1	1	0	1
2	12 Mile Rd W	Orchard Lake Rd	Oakland	87.9	0	17.5	70.4	187.65	5	1	1	1	0	1
3	Van Dyke Ave	11 Mile Rd E	Macomb	87	0	13.6	73.4	164.52	10	1	1	1	0	1
4	Ford Rd	Haggerty Rd	Wayne	82.2	0.1	17.4	64.7	195.91	2	1	1	1	0	1
5	Rochester Rd S	Auburn Rd E	Oakland	81.9	0	14.2	67.7	162.84	12	1	1	1	0	1
6	12 Mile Rd	Dequindre Rd	Oakland	80.6	0.1	17.1	63.4	192.6	3	1	1	1	0	1
7	12 Mile Rd	Dequindre Rd	Macomb	80.6	0.1	17.1	63.4	192.6	3	1	1	1	0	1
8	11 Mile Rd W	Southfield Rd	Oakland	76.4	0.2	11.7	64.5	172.15	8	1	1	1	0	1
9	Gratiot Ave	23 Mile Rd	Macomb	73.6	0.1	15.5	58	176.48	7	1	1	1	1	1
10	14 Mile Rd W	Orchard Lake Rd	Oakland	73.3	0.1	19.1	54.1	196.7	1	0	1	1	1	1
11	Maple Rd W	Orchard Lake Rd	Oakland	71.5	0	16.1	55.4	163.27	11	1	1	1	1	1
12	Highland Rd	Airport Rd	Oakland	69.4	0.1	12.8	56.5	156.89	14	1	1	1	1	1
13	Davison St W	Linwood St	Wayne	67.8	0	14.4	53.4	149.88	16	1	1	1	1	0
14	Woodward Ave E	University Dr	Oakland	67.7	0.1	8.2	59.4	128.97	20	1	1	0	0	1
15	Gratiot Ave	21 Mile Rd	Macomb	65.3	0.1	11.5	53.7	145.38	18	1	1	1	1	1
16	Rochester Rd S	Avon Rd E	Oakland	64.9	0	10.7	54.2	125.89	22	1	1	1	0	1
17	11 Mile Rd	Dequindre Rd	Macomb	60.8	0.1	16.1	44.6	167.1	9	1	1	1	0	0
18	Groesbeck Hwy	15 Mile Rd	Macomb	58	0	16.4	41.6	151.48	15	0	1	1	0	1
19	7 Mile Rd E	Van Dyke St	Wayne	57.3	0.2	12.4	44.7	157.04	13	0	1	0	1	1
20	Ford Rd	Wayne Rd N	Wayne	57	0	12	45	125.4	23	1	1	1	0	1
21	Van Dyke Ave	11 Mile Rd	Macomb	56.7	0.1	10.1	46.5	128.8	21	1	0	1	1	1
22	Allen Rd	Eureka Rd	Wayne	55.9	0.1	13.4	42.4	146.81	17	1	1	1	1	0
23	Gratiot Ave	Conner St	Wayne	54.4	0.2	10.6	43.6	143.88	19	1	1	1	1	1
24	Allen Rd	Northline Rd	Wayne	54.2	0	10.8	43.4	115.76	25	1	1	1	1	1
25	Garfield Rd	Hall Rd	Macomb	53.3	0	10.8	42.5	114.86	26	1	1	1	1	0
26	15 Mile Rd	Ryan Rd	Macomb	53	0	11.4	41.6	117.98	24	1	1	1	1	0
27	14 Mile Rd	Dequindre Rd	Macomb	51.9	0	10.7	41.2	112.89	27	1	1	1	0	1
28	8 Mile Rd W	Haggerty Rd	Wayne	51.8	0	10.5	41.3	111.65	28	1	1	1	1	1
29	Ford Rd	Newburgh Rd N	Wayne	50.8	0	10.4	40.4	110.08	29	1	1	1	1	1
30	Ford Rd	Lilley Rd	Wayne	50.7	0	9.9	40.8	107.13	30	1	1	1	1	1

Table 2. Crash Reduction Factors, Cost and Service Life of Alternatives

Alternatives	Crash Reduction Factors			Capital Cost ($)	O&M Cost ($)	Service Life (Years)
	Fatal	Injury	PDO			
I	0.06	0.05	0.04	20,000	2,000	2
II	0.13	0.11	0.09	35,000	3,500	2
III	0.25	0.23	0.18	80,000	8,000	3
IV	0.30	0.29	0.25	100,000	10,000	4
V	0.46	0.45	0.42	150,000	15,000	4

Table 3. One Year Allocation By Optimization: Available Budget Same As Minimum Improvement

Location	Major Street	Minor Street	County	I	II	III	IV	V	Expected Cost ($)	Benefit ($)
1	Telegraph Rd	12 Mile Rd W	Oakland	0.0	0.0	0.0	0.0	0.0	0	0
2	12 Mile Rd W	Orchard Lake Rd	Oakland	0.0	1.0	0.0	0.0	0.0	35,000	157,830
3	Van Dyke Ave	11 Mile Rd E	Macomb	0.0	0.0	0.0	0.0	0.0	0	0
4	Ford Rd	Haggerty Rd	Wayne	0.0	0.0	0.0	0.0	1.0	150,000	708,677
5	Rochester Rd S	Auburn Rd E	Oakland	0.0	0.0	0.0	0.0	0.0	0	0
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26	15 Mile Rd	Ryan Rd	Macomb	0.0	0.0	0.0	0.0	0.0	0	0
27	14 Mile Rd	Dequindre Rd	Macomb	0.0	0.0	0.0	0.0	0.0	0	0
28	8 Mile Rd W	Haggerty Rd	Wayne	0.0	0.0	0.0	0.0	0.0	0	0
29	Ford Rd	Newburgh Rd N	Wayne	0.0	0.0	0.0	0.0	0.0	0	0
30	Ford Rd	Lilley Rd	Wayne	0.0	0.0	0.0	0.0	0.0	0	0
Total									645,000	2,974,302

Table 4. Allocation Of Alternatives For Base case For A Five Year Planning Period

Intersection	Year	Major Name	Minor Name	Alternatives - Expected Improvements					Total Alternatives	Allocated Cost	O&M cost	Benefit
				I	II	III	IV	V				
1	Year-1	Telegraph Rd	12 Mile Rd W	0	1	0	0	0	1	35,000	0	153,788
2		12 Mile Rd W	Orchard Lake Rd	0	0	0	0	0	0	0	0	0
3		Van Dyke Ave	11 Mile Rd E	0	0	0	0	0	0	0	0	0
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28		8 Mile Rd W	Haggerty Rd	0	1	0	0	0	1	35,000	0	94,004
29		Ford Rd	Newburgh Rd N	0	0	0	0	0	0	0	0	0
30		Ford Rd	Lilley Rd	0	0	0	0	0	0	0	0	0
Sub Total				**1**	**9**	**2**	**0**	**1**	**13**	**645,000**	**0**	**2,327,141**
1	Year-2	Telegraph Rd	12 Mile Rd W	0	0	0	0	0	0	0	3,500	133,795
2		12 Mile Rd W	Orchard Lake Rd	1	0	0	0	0	1	20,000	0	71,216
3		Van Dyke Ave	11 Mile Rd E	0	0	0	0	0	0	0	0	0
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28		8 Mile Rd W	Haggerty Rd	0	0	0	0	0	0	0	3,500	81,784
29		Ford Rd	Newburgh Rd N	0	0	0	0	0	0	0	0	0
30		Ford Rd	Lilley Rd	0	0	0	0	0	0	0	0	0
Sub Total				**2**	**3**	**0**	**0**	**2**	**7**	**445,000**	**64,500**	**3,720,826**
1	Year-3	Telegraph Rd	12 Mile Rd W	0	0	0	0	0	0	0	0	0
2		12 Mile Rd W	Orchard Lake Rd	0	0	0	0	0	0	0	2,000	66,943
3		Van Dyke Ave	11 Mile Rd E	0	0	1	0	0	1	80,000	0	280,378
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28		8 Mile Rd W	Haggerty Rd	0	0	0	0	0	0	0	0	0
29		Ford Rd	Newburgh Rd N	0	0	0	0	0	0	0	0	0
30		Ford Rd	Lilley Rd	0	0	0	0	0	0	0	0	0
Sub Total				**1**	**1**	**1**	**0**	**4**	**7**	**735,000**	**75,500**	**4,336,179**
1	Year-4	Telegraph Rd	12 Mile Rd W	0	0	0	0	0	0	0	0	0
2		12 Mile Rd W	Orchard Lake Rd	0	0	0	0	0	0	0	0	0
3		Van Dyke Ave	11 Mile Rd	0	0	0	0	0	0	0	8,000	210,283
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28		8 Mile Rd W	Haggerty Rd	0	0	0	0	0	0	0	0	0

29		Ford Rd	Newburgh Rd N	0	0	0	0	0	0	0	0	0
30		Ford Rd	Lilley Rd	0	0	0	0	0	0	0	0	0
Sub Total				**3**	**2**	**1**	**2**	**0**	**8**	**410,000**	**118,500**	**4,276,215**
1	Year-5	Telegraph Rd	12 Mile Rd W	0	0	0	0	0	0	80,000	0	275,295
2		12 Mile Rd W	Orchard Lake Rd	0	0	1	0	0	1	35,000	0	148,360
3		Van Dyke Ave	11 Mile Rd E	0	0	0	0	0	0	0	8,000	157,483
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28		8 Mile Rd W	Haggerty Rd	0	0	0	0	0	0	0	0	0
29		Ford Rd	Newburgh Rd N	0	0	0	0	0	0	100,000	0	248,700
30		Ford Rd	Lilley Rd	0	0	0	1	0	1	0	0	0
Sub Total				**1**	**1**	**2**	**2**	**2**	**8**	**715,000**	**139,000**	**3,637,375**

Table 5. Optimization Model Results for Base case and other applied cases

Model	Year	Number of Alternatives Allocated						Benefit ($)	Allocated ($)	O&M Cost ($)	Budget ($)	Surplus ($)	Cumulative ($)
		I	II	III	IV	V	Total						
Base case (a)	1	1	9	2	0	1	13	2,327,141	645,000	0	645,000	0	0
	2	2	3	0	0	2	7	3,720,826	445,000	64,500	645,000	135,500	135,500
	3	1	1	1	0	4	7	4,336,179	735,000	75,500	677,250	-133,250	2,250
	4	3	2	1	2	0	8	4,276,215	410,000	118,500	677,250	148,750	151,000
	5	1	1	2	2	2	8	3,637,375	715,000	139,000	711,113	-142,888	8,113
	Total	**8**	**16**	**6**	**4**	**9**	**43**	**18,297,736***	**2,950,000**	**397,500**	**3,355,613**	**8,113**	
Relative urgency case (b)	1	0	4	3	0	2	9	2,586,154	680,000	$0	645,000	-35,000	-35,000
	2	0	3	2	0	1	6	4,319,017	415,000	68,000	645,000	162,000	127,000
	3	0	2	1	0	2	5	3,309,989	450,000	95,500	677,250	131,750	258,750
	4	0	1	0	1	3	5	3,554,997	585,000	106,000	677,250	-13,750	245,000
	5	2	3	3	0	3	11	4,140,809	835,000	111,500	711,113	-235,388	$9,613
	Total	**2**	**13**	**9**	**1**	**11**	**36**	**17,910,966***	**2,965,000**	**381,000**	**3,355,613**	**9,613**	
County constraint case (c)	1	7	5	2	0	0	14	1,663,429	475,000	$0	645,000	170,000	170,000
	2	1	1	3	0	2	7	3,574,647	595,000	47,500	645,000	2,500	172,500
	3	1	4	2	0	2	9	4,122,963	620,000	59,500	677,250	-2,250	170,250
	4	3	1	2	2	1	9	4,054,100	605,000	62,000	677,250	10,250	180,500
	5	0	5	3	1	2	11	4,111,661	815,000	60,500	711,113	-164,388	16,113
	Total	**12**	**16**	**12**	**3**	**7**	**50**	**17,526,800***	**3,110,000**	**229,500**	**3,355,613**	**16,113**	
Relaxation of carry -over feature case (d)**	1	3	3	1	1	2	10	2,433,053	645,000	0	645,000	0	0
	2	5	5	0	0	2	12	4,729,963	575,000	64,500	645,000	5,500	5,500
	3	5	5	1	1	1	13	4,342,156	605,000	72,500	677,250	-250	5,250
	4	7	9	2	0	0	18	4,082,024	615,000	50,000	677,250	12,250	17,500
	5	5	6	1	0	2	14	4,347,805	690,000	30,000	711,113	-8,888	8,613
	Total	**25**	**28**	**5**	**2**	**7**	**67**	**19,935,001***	**3,130,000**	**217,000**	**3,355,613**	**8,613**	

Note: Assumption- alternatives are mutually exclusive

* : Total Benefit is the objective function of the optimization model

**: Scenario-3 the mutually exclusive alternative feature is relaxed

Table 6. Optimization Model Results for Base case and applied cases

Model	Year	New				Carry-Over				Total			
		Wayne	Oakland	Macomb	Total	Wayne	Oakland	Macomb	Total	Wayne	Oakland	Macomb	Total
Base case (a)	1	6	3	4	13	0	0	0	0	6	3	4	13
	2	1	3	3	7	6	3	4	13	7	6	7	20
	3	1	4	2	7	4	3	3	10	5	7	5	17
	4	3	2	3	8	2	5	3	10	5	7	6	18
	5	1	4	3	8	4	6	5	15	5	10	8	23
	Total	**12**	**16**	**15**	**43**	**16**	**17**	**15**	**48**	**28**	**33**	**30**	**91**
Relative urgency case (b)	1	3	3	3	9	0	0	0	0	3	3	3	9
	2	0	5	1	6	3	3	3	14	3	8	4	15
	3	2	1	2	5	2	7	2	9	4	8	4	16
	4	0	2	3	5	3	5	2	14	3	7	5	15
	5	2	5	4	11	2	3	4	15	4	8	8	20
	Total	**7**	**16**	**13**	**36**	**14**	**20**	**18**	**52**	**17**	**34**	**24**	**75**
County constraint case (c)	1	5	5	4	14	0	0	0	0	5	5	4	14
	2	2	2	3	7	5	5	4	14	7	7	7	21
	3	3	3	3	9	3	3	3	9	6	6	6	18
	4	2	4	3	9	4	5	5	14	6	9	8	23
	5	5	3	3	11	2	7	6	15	7	10	9	26
	Total	**17**	**17**	**16**	**50**	**14**	**20**	**18**	**52**	**31**	**37**	**34**	**102**